Jean-Guillaume de Damas

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Experience

CERMICS, CNRS, ENPC, Institut Polytechnique de Paris, Postdoctoral Researcher — Champs-sur-Marne, France
May 2026 – present
7 months

- Working on hybrid methods combining model reduction and machine learning with Virginie Ehrlacher, at the Centre d'Enseignement et de Recherche en Mathématiques et Calcul Scientifique
- Developing numerical methods for accelerating parametric studies in crowd and granular media simulations
- Implementing algorithms in the SCOPI++ software, in collaboration with Aline Lefebvre-Lepot (ENS Paris-Saclay) and Loïc Gouarin (CMAP)

CERFACS, Master's Intern – Researcher & Scientific Programmer — Toulouse, France
Apr 2022 – Oct 2022
7 months

- Researched randomized SVD preconditioners for conjugate gradient methods, with applications to data assimilation
- Delivered functional C++ code for the European Centre for Medium-Range Weather Forecasts (ECMWF)
- Supervised by Selime Gürol

Education

PhD **INRIA Paris & Sorbonne Université**, Applied Mathematics — Paris, France
Nov 2022 – Mar 2026

- Thesis: Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations
- Designed, implemented, and theoretically analyzed new methods for linear algebra routines based on dimension reduction via randomized projection, with an HPC focus
- Achieved speed-ups of up to 3–4× over state-of-the-art methods while preserving accuracy
- Supervised by Laura Grigori, within the ERC project EMC2

MSc **ISAE-SUPAERO**, Engineering — Toulouse, France
Sept 2018 – Nov 2022

- Machine learning: algorithms for SVMs, neural networks, reinforcement learning, and mathematical foundations
- Statistics: Bayesian inference, stochastic algorithms, non-parametric estimation
- Fluid dynamics: theory, computational aspects, wind tunnel experiments
- Computer science: high-performance computing with parallelization in C, functional programming languages
- Optimization: inverse problems, data assimilation, regularization, and numerical challenges in finite-precision arithmetic

Publications

Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations — June 2026
Jean-Guillaume de Damas
10.70675/42a47550z0087z4fddzbf44z13273ef37f8b (PhD Thesis)

Randomized Implicitly Restarted Arnoldi Method for the Non-Symmetric Eigenvalue Problem — Oct 2025
Vol. 46, No. 4, pp. 2395–2422. ISSN: 1095-7162.
Jean-Guillaume de Damas, Laura Grigori
10.1137/24m1674352 (SIAM Journal on Matrix Analysis and Applications)

Randomized Krylov-Schur Eigensolver with Deflation Aug 2025
Jean-Guillaume de Damas, Laura Grigori
10.48550/ARXIV.2508.05400 (arXiv)

Randomized Orthogonalization and Krylov Subspace Methods: Principles and Algorithms Dec 2025
Jean-Guillaume de Damas, Laura Grigori, Igor Simunec, Edouard Timsit
10.48550/arXiv.2512.15455 (arXiv)

Selected Talks

Workshop on Approximate Computing in Numerical Linear Algebra Oct 2025
A randomized Krylov-Schur eigensolver, featuring mixed-precision and deflation

CERMICS Applied Mathematics Seminar June 2025
Randomized methods for the eigenvalue problem and low rank approximation

SIAM Conference on Applied Linear Algebra LA24 & CERFACS Sparse Days May 2024
Randomized Implicitly Restarted Arnoldi algorithm for the unsymmetric eigenvalue problem

Teaching

Analysis and Partial Differential Equations, Part II (24 h) Spring 2027
École nationale des ponts et chaussées (ENPC), Champs-sur-Marne

Analysis and Partial Differential Equations, Part I (24 h) Winter 2026
École nationale des ponts et chaussées (ENPC), Champs-sur-Marne

Mathematics for Science Studies I (52 h) Fall 2023
Sorbonne University, Paris

Programming Basics I (20 h) Fall 2023
Sorbonne University, Paris

Skills

Programming
Julia, Python, MATLAB, R, C, C++

Technologies
PyTorch, Scikit-learn, OpenMP, MPI, CUDA

Expertise
Numerical Analysis, Randomized Linear Algebra, Eigensolvers, Low-Rank Approximations

Languages

English
Advanced (C1, TOEFL ITP 600/677)

German
Intermediate (A2)

Japanese
Beginner (A1)

Outside of the office

- Piano & musicology: learning as a passionate amateur with conservatory-level lessons, and composing when time allows
- Video game development: building a custom engine in C++, or using Unity/Unreal